EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2026-27

Course: Wireless and Mobile Communication **Course code:**
Year: TE **SEM: V**

Experiment No. 7
Aim: To study Bluetooth communication between mobile devices by performing device pairing and testing the communication range.
Name: Meet Mehta
Roll Number: 30
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	5	3	2
Understanding	5	3	2
Journal work and timely submission.	10	8	4

Checked by
Name of Faculty : Mrs. Sonali Sable
Signature : **Date :**

Aim

To study Bluetooth communication between mobile devices by performing device pairing and testing the communication range.

Theory

1. Bluetooth Communication

Bluetooth is a short-range wireless communication technology used to exchange data between nearby electronic devices without using physical cables.

It can be used for communication between devices such as smartphones, laptops, wireless headphones, speakers, keyboards, and other Bluetooth-enabled devices.

2. Bluetooth Pairing

Bluetooth pairing is the process of establishing a trusted relationship between two Bluetooth-enabled devices.

During pairing, one device searches for nearby Bluetooth devices and identifies the required device by its Bluetooth name.

A pairing request and verification code may be displayed on both devices. The devices can be paired after confirming the displayed code.

3. Bluetooth Connection

After two devices are successfully paired, a Bluetooth connection can be established when required by the communication application or selected data-transfer method.

Once connected, the devices can exchange supported data such as files, photos, or other information.

4. Bluetooth Communication Range

Bluetooth is designed for short-range wireless communication.

The actual communication range depends on factors such as:

- **Bluetooth device capability**
- **Transmit power**
- **Distance between devices**
- **Obstacles and walls**
- **Device orientation**
- **Surrounding radio environment**

Therefore, the maximum reliable range may vary under different experimental conditions.

5. Effect of Distance on Bluetooth Communication

As the distance between two Bluetooth devices increases, the received signal generally becomes weaker.

At shorter distances, communication is usually more reliable. As the devices are moved farther apart, communication may become unstable or eventually fail, depending on the devices and environment.

6. Effect of Obstacles

Walls and other physical obstacles can affect Bluetooth communication.

Therefore, the communication performance obtained in an open environment may differ from that obtained in an indoor environment with walls or other obstacles.

Procedure

Step 1: Select Two Bluetooth-Enabled Smartphones

Take two smartphones having Bluetooth communication capability.

For identification, consider them as Smartphone 1 and Smartphone 2.

Step 2: Open Bluetooth Settings

Open the Settings application on both smartphones.

Navigate to the Bluetooth settings.

Step 3: Enable Bluetooth

Turn Bluetooth ON on both smartphones.

Step 4: Enable Device Visibility

Open the Bluetooth settings and make the required device discoverable/visible, if this option is available or required by the smartphone.

Step 5: Scan for Nearby Devices

On Smartphone 1, select Pair New Device, Scan for Devices, or the corresponding option provided by the smartphone.

Step 6: Identify Smartphone 2

Wait for the list of nearby Bluetooth devices to appear.

Identify the Bluetooth name of Smartphone 2 from the displayed list.

Step 7: Initiate Pairing

Select Smartphone 2 from the list to initiate the Bluetooth pairing process.

Step 8: Observe the Pairing Request

Observe the Bluetooth pairing request and pairing code displayed on both smartphones.

Step 9: Verify the Pairing Code

Check whether the pairing code displayed on both smartphones is the same.

If the codes match, select Pair/Confirm on both smartphones.

Step 10: Verify Pairing

Verify that Smartphone 2 appears in the Paired Devices list of Smartphone 1.

Similarly, verify that Smartphone 1 appears as a paired device on Smartphone 2.

Step 11: Establish Bluetooth Communication

Establish an active Bluetooth connection between the two smartphones, if required by the selected communication method.

Step 12: Perform Data Transfer

Send a small file, photo, or other supported data from one smartphone to the other to verify Bluetooth communication.

Step

13: Record Communication Status

Observe whether the data transfer is:

- Successful
- Unstable
- Unsuccessful

Record the result in the observation table.

Step 14: Perform Range Testing

Initially place the two smartphones approximately 1 m apart.

Perform the same Bluetooth communication test and record the communication status.

Step 15: Increase the Distance

Gradually increase the distance between the smartphones, for example:

3 m → 5 m → 10 m → 15 m → 20 m

At each distance, perform the same communication test.

Step 16: Record Data Transfer

At each distance, record whether the data transfer is:

- Successful
- Unstable
- Failed

Step

17: Test Indoor Environment

If possible, repeat the range test in an indoor environment with walls or other obstacles.

Observe any difference in communication performance.

Step 18: Compare the Results

Compare the Bluetooth communication performance at different distances and under different environmental conditions.

Observe how increasing distance and obstacles affect communication.

Step 19: Determine Maximum Reliable Range

Identify the maximum distance at which reliable Bluetooth communication is obtained under the experimental conditions.

Step 20: Record Observations and Conclusion

Record all observations in the observation tables and write the conclusion based on the Bluetooth pairing and range-testing results.

Observation Table – Bluetooth Pairing

Sr. No.	Device Name	Bluetooth Status	Pairing Status	Pairing Code Verified	Communication Status
1	Smartphone 1	ON	Paired	Yes	Successful

Step					
2	Smartphone 2	ON	Paired	Yes	Successful

Observation Table – Bluetooth Range Testing

Sr. No.	Distance (m)	Environment	Communication Status	Data Transfer	Observation
1	1	Open/Indoor	Successful	Successful	Good Communication
2	3	Open/Indoor	Successful	Successful	Stable Communication
3	5	Open/Indoor	Successful	Successful	Stable Communication
4	10	Open/Indoor	Successful	Successful	Good Communication
5	15	Open/Indoor	Successful	Successful	Stable Communication
6	20	Open/Indoor	Unsuccessful	Failed	Bluetooth communication was not su

Range Analysis:

Distance Range	Communication Quality	Observation
Short distance	Excellent	Communication and data transfer were successful and stable.
Medium distance	Good	Communication remained successful with stable data transfer.
Long distance	Poor/Unreliable	Communication started becoming unreliable and failed at 20 m.
Maximum reliable range	Up to 15 m	Successful communication and data transfer were observed up to 15 m.

Conclusion:
Bluetooth communication was successfully established between the smartphones at shorter distances. Data transfer remained stable up to 15 m, while communication failed at 20 m. Thus, the observed maximum reliable Bluetooth range in this experiment was approximately 15 m.

Viva Questions:

1. What is Bluetooth communication?
Bluetooth communication is a short-range wireless technology used to exchange data between nearby devices without cables.

2. What is Bluetooth pairing?
Bluetooth pairing is the process of establishing a secure connection between two Bluetooth-enabled devices.

3. Why is the pairing code verified on both smartphones?
The pairing code is verified to confirm that the correct devices are being connected and to prevent unauthorized connections.

4. What happens to Bluetooth communication as the distance between devices increases?
As the distance increases, the Bluetooth signal strength decreases, causing slower or unstable communication and eventually connection failure.

5. What factors affect Bluetooth communication range?
The range is affected by distance, obstacles/walls, interference from other wireless devices, signal strength, device hardware, and the Bluetooth version/class.